

A Review of Ji-Hyun Song's and So-Yeon
Kim's Article

Push-Pull Mechanism of Attention and Emotion in Children with ADHD

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Introduction

Introduction- Background

What is ADHD?

- Neurodevelopmental disorder with main symptoms being inattention, hyperactivity, and impulsivity
- Core feature of ADHD is a deficit in attentional operations → influences the process of efficiently picking relevant information in an individual's surroundings to accomplish goals

Introduction- Previous Studies

- Posner- outlined three attentional networks
 - 1. Alerting network- maintains vigilance
 - 2. Orienting network- manages rapid shifting of attention → less clear in relation to ADHD
 - 3. Executive control network- responsible for selective attention and conflict resolution
- Ortega and Colleagues
 - Reported abnormal ERP components linked with attentional orienting in ADHD
- Kim et al.
 - ADHD also linked to deficits in socio-emotional functions, including challenges in recognizing negative facial emotions, making social interactions more difficult
 - Faces can involuntarily capture attention, causing attentional biases, especially toward threatening facial stimuli

Aims and Hypotheses

Aims and Hypotheses- Bridging the Gap

- There is little research specifically investigating attentional bias towards facial stimuli among children with ADHD
- Existing research overlooks an individual's anxiety levels despite 25-30% of children with ADHD having anxiety symptoms
- This is important to study because anxiety has the potential to exacerbate ADHD symptoms and disrupt emotional interference control → can cause challenges in dealing with emotional stimuli

Aims and Hypotheses

- This study aims to investigate the relationship between attentional bias towards emotional facial stimuli and anxiety levels in children with ADHD, in comparison with TD children
- Examined impacts of anxiety levels on attentional bias effects toward faces in both TD and ADHD children
- In two experiments, investigated if children with ADHD demonstrate atypical attentional-orienting functions, particularly with a deficiency in attentional capture effects towards threatening stimuli
- Additionally, examined whether these atypical orienting functions in ADHD are modulated by anxiety levels

Aims and Hypotheses- Research Questions

- Do children with ADHD exhibit atypical attentional-orienting functions, indicated by a lack of attentional capture effects towards threatening stimuli?
- Are the atypical orienting functions in children with ADHD modulated by their anxiety levels?

Aims and Hypotheses

- **Hypothesis 1.** Children with ADHD will display deficits in attentional bias towards emotional facial stimuli compared to TDC
- **Hypothesis 2.** Higher levels of trait anxiety in children with ADHD will be positively associated with an attentional bias towards threatening stimuli

Methods- Experiment 1

Methods- Experiment 1

Continuous Performance Task (CPT)- Task Directions

- Target stimulus = red “T” overlapping a black cross, appeared at top right of screen
- Participants instructed to fixate on a central point while distractors appeared for 4 seconds in the center of the screen
- Ignore distractor stimuli and respond to direction of the target (red “T”)
- If target stimulus appeared vertically or horizontally (0, 90, 180, 270 degrees) participants pressed the left mouse button with right hand
- If target stimulus appeared diagonally (45, 135, 225, 315 degrees) participants pressed the right mouse button with right hand

Methods- Experiment 1

2 types of conditions for each group- distractor type and time condition

- Distractor type (2 levels)
 - Angry Face
 - Neutral Place
- Time condition (5 levels)
 - T1- indicated that target presented concurrently with the distractor for 1000 ms
 - T2- indicated the target presented 1000 ms after T1
 - T3- indicated the target presented 1000 ms after T2
 - T4- indicated the target presented 1000 ms after T3
 - TBaseline (TB)- represented trials where only the target was presented without distractors and was randomly presented within four conditions lasting 3000 to 6000 ms (reference RT)

Methods- Experiment 1

- Attentional capture = slow reaction time to target at T1
- Attentional holding = slow response time to target from T2 to T4

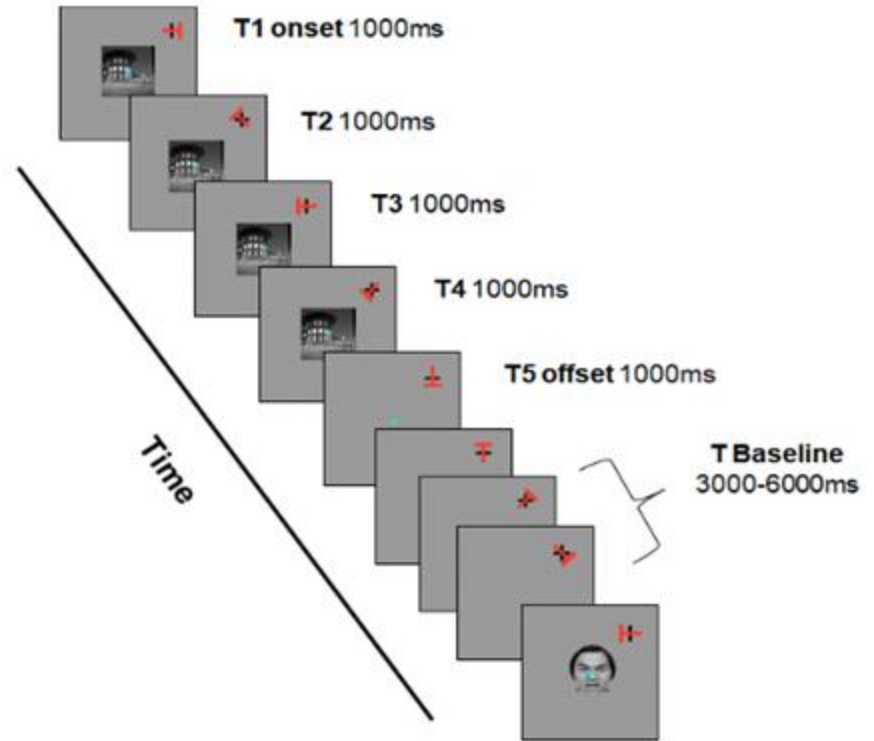


Figure 1. The attention task paradigm of Experiment 1.

Pre-Experiment

State-Trait Anxiety Inventory: STAI-X I,II and Pre-CPT

- Prior to the experiment, participants completed the STAI-X I and II questionnaires.
 - Used in this study to measure levels of state anxiety and trait anxiety
 - Each anxiety scale contained 20 items, rated on a scale from 1 - 4, scores ranged from 20 - 80 (higher scores represent higher levels of state and trait anxiety)
- Prior to performing the main CPT, all participants took part in a passive viewing task intended to expose them to stimuli used in the primary task, to control for possible attentional capture effects from novel stimuli

Results- Experiment 1

Statistical Analysis

Outliers (Same for Both Experiments)

- Reaction times (RTs) less than 150 ms and greater than 1150 ms in the attention task were excluded from the analysis based on previous research
- RT of 150 ms is too fast for attention to be properly directed in the current task paradigm
- Target changed its direction every 1000 ms → unclear which trial correlates to a RT greater than 1150 ms
- For accuracy and RTs, participants with mean data 3 standard deviations away from group average → determined as outliers and excluded from the final analyses.

Statistical Analysis

Overview

- Three-way repeated-measure ANOVA to evaluate effects of 3 main factors on (RTs)
 - Distractor type- whether distraction was angry face or neutral place
 - Time order- time at which target and distractor appeared
 - Group- the 2 groups of participants (ADHD of TD children)
- Purpose of ANOVA- see if these three factors and interactions between them had significant effect in the way children responded to task.

Statistical Analysis

Main Effects

- Time Order
 - Analysis demonstrated significant main effect of time order
 - RTs varied significantly based on when the target appeared relative to the distractor (at T1,2,3,4,or TB)
 - Distractors (angry faces neutral places) had different effects on RTs at different point in time
- How Time Order Interacts with Group
 - Significant interaction between time order and group → effect of time order differed between ADHD and TDC groups
 - RTs of two groups changed differently across T1-T4 based on distractor

Post Hoc Analysis- TDC

- Post hoc paired t-tests conducted to compare RTs for different distractor types and time orders for ADHD children and TDC separately

Result of Post Hoc Analysis for TDC

- **Attentional Capture**
 - TDC demonstrated attentional capture effects for angry faces and neutral place distractors → when target presented at same time as distractor (T1), RTs were significantly slower
- **Attentional Holding**
 - Place Images- TDC also displayed significant attentional holding effect at T2 → even after the distractor is presented for a period of time, RTs still slower → sustained distraction effect from place stimuli
 - Face Images- TDC did NOT display attentional holding for angry faces following T1 → fast recovery from distraction created by angry faces

Post Hoc Analysis- ADHD

Result of Post Hoc Analysis for ADHD Children

- **Attentional Capture**

- ADHD group did NOT demonstrate significant attentional capture effect at T1 for either type of stimuli → not significantly distracted by angry faces or neutral places

- **Attentional Holding**

- ADHD group did NOT display attentional holding effects → RTs were not significantly slowed following initial distractor

- **Less affected by distractors**

ANOVA- Experiment 1

Table 2. An ANOVA table in Experiment 1.

	<i>F</i>	<i>p</i>	η^2
Distractor	$F(1, 44) = 2.41$	0.13	0.05
Time order	$F(2.98, 131.10) = 18.67$	<0.001	0.30
Group	$F(1, 44) = 1.84$	0.18	0.04
Distractor $\times$ Group	$F(1, 44) = 0.51$	0.48	0.01
Time order $\times$ Group	$F(2.98, 131.10) = 9.20$	<0.001	0.17
Distractor $\times$ Time order	$F(4, 176) = 0.786$	0.54	0.02
Distractor $\times$ Time order $\times$ Group	$F(3.51, 154.46) = 2.18$	0.083	0.05

Correlation Analysis

- Researchers conducted correlation analysis to see whether there was a relationship between levels of state or trait anxiety and RTs to distractors
- Results indicated no significant correlation between distraction effects and anxiety levels in both ADHD and TD children

Attentional Bias Effect on Distractor Type & Time Order

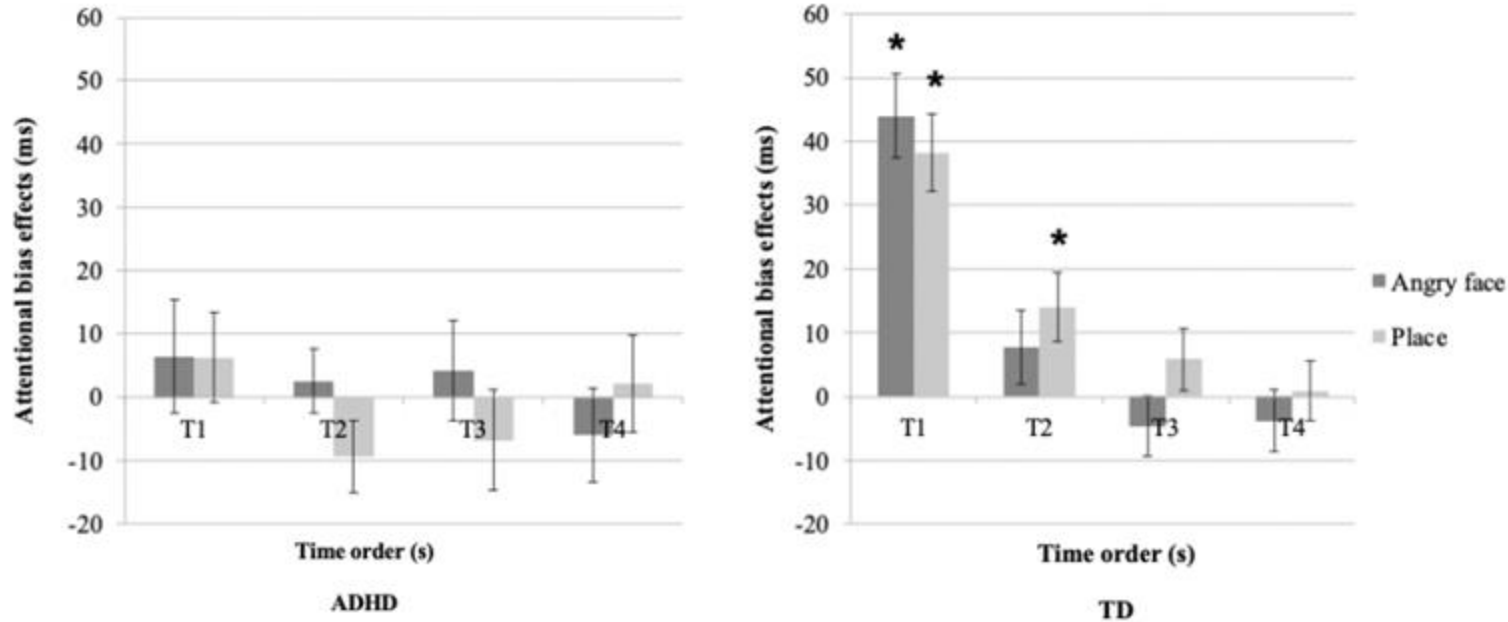


Figure 2. Attentional bias effect on the distractor type and time order in Experiment 1 (* indicates a significant effect with $p < 0.05$).

Takeaways from Experiment 1

- Children with ADHD were less likely to be distracted by both emotional and non-emotional stimuli (angry faces and neutral places) when compared to TDC
- TDC were more distracted by both types of stimuli, particularly by neutral places as these types of stimuli sustained their attention for a longer period of time than images of angry faces

Methods- Experiment 2

Methods- Experiment 2

Participants

- 21 children aged 11 to 15 diagnosed with ADHD (17 boys and 4 girls)
- 25 age-matched TD children (16 boys and 9 girls)
- All had normal or corrected-to-normal vision and were right handed

Group	Sex		Age	State Anxiety	Trait Anxiety
	Boy (%)	Girl (%)	M (SD)	M (SD)	M (SD)
ADHD (N = 21)	17 (81)	4 (19)	12.64 (1.29)	38.14 (7.84)	49.14 (9.54)
TDC (N = 25)	16 (64)	9 (36)	12.64 (1.07)	35.24 (5.39)	41.48 (9.14)

Methods- Experiment 2

- Experiment 2 used same Continuous Performance Tasks as experiment 1 except instead of non-emotional place images, included non-emotional neutral face stimuli + angry face stimuli
- 4 women and 4 men in each set of face stimuli totaling 16 face stimuli
- Other than the types of distractors, Experiment 2's procedures, anxiety questionnaires, and statistical analyses are the same as those in Experiment 1

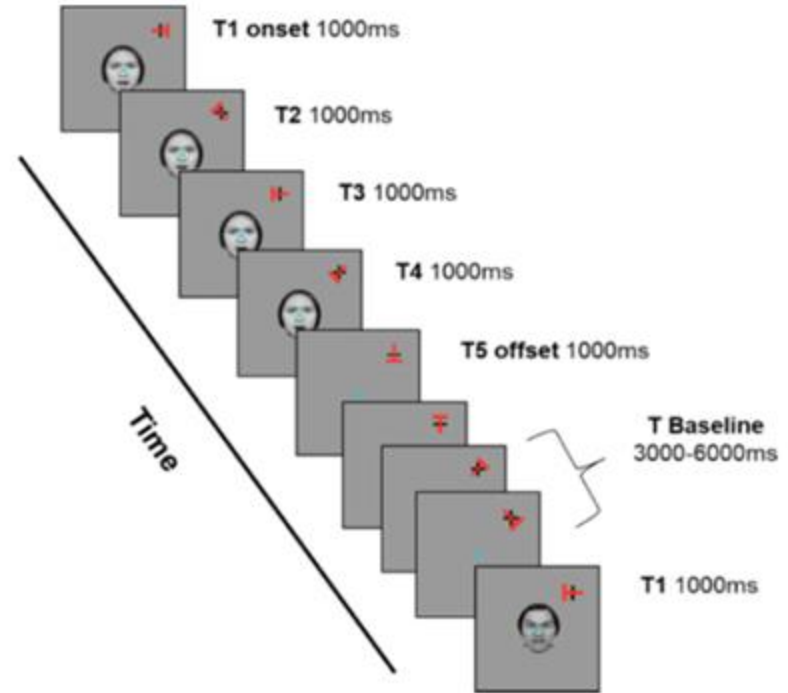


Figure 3. The attention task paradigm of Experiment 2.

Results- Experiment 2

Results- Experiment 2

- Three-way mixed ANOVA (2 (Group: ADHD/TDC) $\times$ 2 (distractor type: angry/neutral face) $\times$ 5 (time order: T1, T2, T3, T4, TB) carried out on RT data in CPT indicated information about-
 - Main Effect of Distractor Type
 - Main Effect of Time Order
 - Attention Capture in ADHD Children
 - Attention Capture in TDC
 - Correlation with Anxiety Levels

Main Effects- Experiment 2

Main Effect of Distractor Type

- Trend-level significance for differences in attentional bias between angry and neutral face stimuli ($p = 0.054$)
 - → angry faces as distractors potentially have stronger influence on attentional capture in comparison to neutral faces; p-value suggest only marginally significant

Main Effect of Time Order

- Significant variation in RTs across different points in time ($p < 0.001$)
 - → RTs varied based on when target appeared relative to distractor

Attention Capture- Experiment 2

Attention Capture in ADHD Children

- Children with ADHD displayed significant attentional capture effect for angry face distractors at T1 ($p < 0.05$)
 - → were more distracted by angry faces when target appeared right after distractor
 - → did not show similar effect for neutral face distractors

Attention Capture in TDC

- TDC displayed attentional capture for angry face and neutral face distractors at T1 ($p < 0.001$)
 - → more generally impacted by presence of any face stimuli, no matter the emotional content

ANOVA Experiment 2

Table 4. An ANOVA table in Experiment 2.

	<i>F</i>	<i>p</i>	η^2
Distractor	$F(1, 44) = 3.91$	0.05	0.08
Time order	$F(2.97, 130.85) = 11.83$	<0.001	0.21
Group	$F(1, 44) = 0.13$	0.72	0.003
Distractor $\times$ Group	$F(1, 44) = 0.97$	0.33	0.02
Time order $\times$ Group	$F(4, 176) = 0.997$	0.41	0.02
Distractor $\times$ Time order	$F(2.81, 123.63) = 0.75$	0.56	0.02
Distractor $\times$ Time order $\times$ Group	$F(2.81, 123.63) = 0.38$	0.77	0.01

Attentional Bias Effect on Distractor Type & Time Order

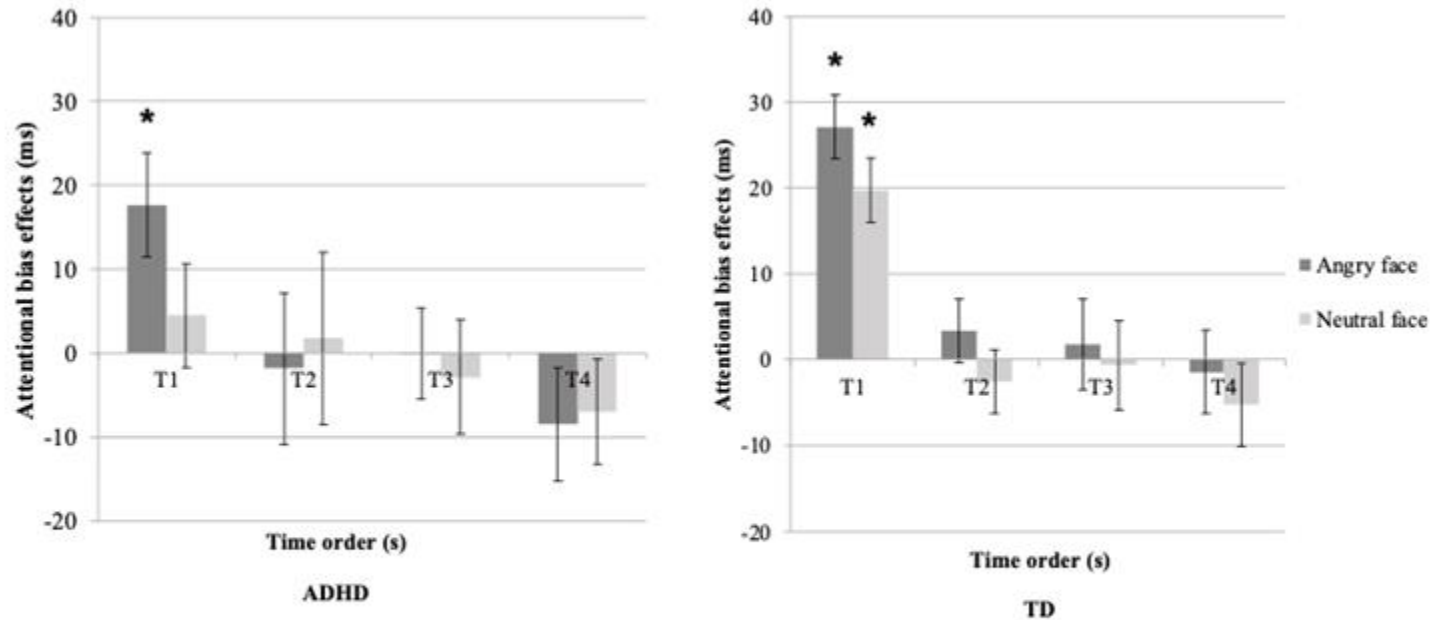


Figure 4. Attentional bias effect on the distractor type and time order in Experiment 2. (* indicates a significant effect with $p < 0.05$).

Correlation with Anxiety Levels– Experiment 2

- Significant positive correlation ($r = 0.61$, $p < 0.05$) revealed between trait anxiety levels and attentional capture from angry face distractors in ADHD group
 - → children with greater levels of trait anxiety = more prone to be distracted by angry faces

Correlation with Anxiety Levels– Experiment 2

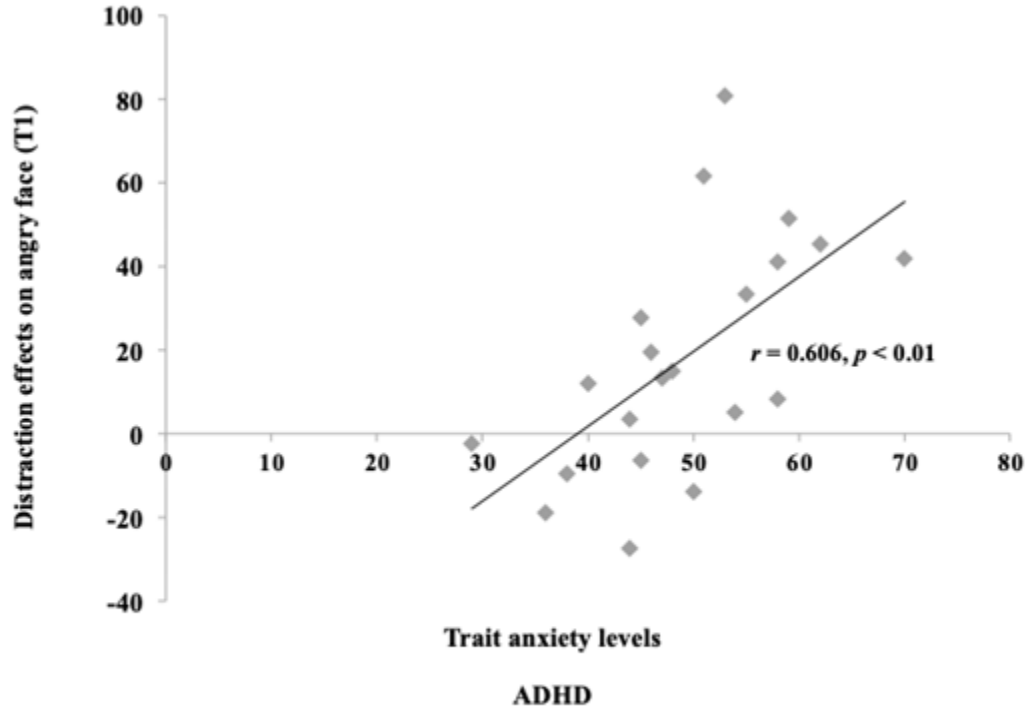


Figure 5. Correlation between trait anxiety and distraction effects on angry face in ADHD children in Experiment 2.

Takeaways from Experiment 2

- Correlation ($r = 0.61$) and associated p-value ($p < 0.05$) indicates → as trait anxiety levels increase, ADHD children tend to display higher attentional capture from threatening stimuli
- On the other hand, TDC did not exhibit this particular relationship between anxiety and attentional capture → suggesting anxiety levels modulate attention differently between the two groups of children
- Emotional stimuli, especially those perceived as threatening, create greater impact on attention in anxious children with ADHD

Discussion

Discussion– Main Findings of Overall Study

- Atypical Attentional Bias in Children with ADHD
 - In Experiment 1, children with ADHD showed no attentional capture for both angry face and neutral place stimuli, suggesting deficits in capacity to orient attention to these stimuli
 - In Experiment 2, children with ADHD showed attentional capture only for angry face distractors, demonstrating a bias towards emotionally threatening stimuli, not neutral stimuli
- How Anxiety Plays a Role
 - In Experiment 2 → significant positive correlation revealed between trait anxiety levels and attentional capture from angry faces in children with ADHD
 - Suggests that children with greater levels of anxiety are more likely to experience attentional capture from threatening stimuli
 - Anxiety did not modulate attentional bias in TDC, demonstrating that anxiety uniquely influences attentional processes in ADHD children

Discussion– Main Findings of Overall Study

Differences Displayed Between ADHD Children and TDC

- In Experiment 2, TDC demonstrated attentional capture for both angry and neutral face stimuli; Children with ADHD only responded to angry face stimuli
 - → indicate that TDC have typical attentional responses to visual stimuli, no matter what the emotional content is; ADHD children's attention rather emotionally driven , especially impacted by levels of anxiety

Thank you!